V1.3 — Reading Real Language Safely

Reflex Engine, experiment V1.3 · OUTCOME D · 2026-08-31

The engine answers questions written in ordinary English, and it does so at a measured wrong-answer rate rather than a promised one. Its best arm reads 69.7% of real language with three wrong answers in 209, and a separate retrieval track certifies 289 answers with none wrong.

And the pre-registered verdict is D, not A, because the weakest arm came in at 2.63% — above the 2% gate — on a single wrong answer in 38. The ladder was frozen before any sealed evaluation and puts D above A. Section 2 explains why that reading is honoured rather than argued around.

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1. The question this version had to answer

V1.2 closed the generator universe and then hit a wall stated plainly in its own report: on real user language — GSM8K, ASDiv, SciQ, frozen and sealed — the engine abstained on **519 of 520** items. Coverage was, to a rounding error, zero. Two acquisition loops run against that residual produced **zero lift on both arms**. The diagnosis was that the gap was not vocabulary but *arithmetic story composition*.

V1.3 acts on that diagnosis with one structural change, plus the two extensions the user asked for:

> *"i want it to work with normal human language and also make it self generating and not just arithmetic"*

- **Normal human language** → the parse/verify split (§3).
- **Not just arithmetic** → provenance certification for retrieved facts, which involve no computation at all (§5).
- **Self-generating** → plan reflexes, minted from the engine's own verified solves (§6). This is the one that failed.

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2. Verdict against the frozen ladder

The outcome ladder was fixed in `V1_3_SPEC.md` before any sealed evaluation, with priority **F > D > C > B > A**.

	condition	result
F	PARITY-PLUS broken by W1–W4 changes	**not triggered** — Tier A re-verified live

	condition	result
D	any sealed suite above the 2% gate, or Tier A regressed	**TRIGGERED** — haiku arm 1/38 = **2.63%**
C	L3 fails	not reached — D outranks
B	L3 + L9, L5 null *and* L7/L8 partial	not reached — D outranks
A	L3 + L9 + (L5 **or** L7/L8)	not reached — D outranks

OUTCOME D. The haiku arm produced one wrong answer in 38 certified = **2.63%**, above the pre-registered 2% gate.

Why this is reported as D

Every other reading of the run says A. The strongest arm is at 1.44%, the two clean arms combined are at 1.62%, the retrieval track is at 0.00%, and Tier A is intact. If the gate were applied to the aggregate, or to the arm anyone would deploy, the version passes.

None of that is the pre-registration. The ladder says *"any sealed suite or sweep above the 2% gate"*, with the explicit rider that **"a safety regression outranks every coverage claim"**, and it fixes the priority F > D > C > B > A so that D is tested before A. The haiku arm is a sealed measurement and it is above the gate.

Choosing, after seeing the data, to read "suite" as excluding the arm that failed — or to substitute the combined figure for the per-arm one — would be exactly the move this arc pre-registers to prevent. The verdict is D.

What D means here, stated precisely: it is one wrong answer on the smallest and weakest arm, at n=38, where the 95% interval runs [0.47%, 13.50%] and no honest claim can be made in either direction. It is not evidence that the approach is unsafe, and it is not dismissible either. It means the version did not demonstrate the safety margin it set out to demonstrate, on every arm it measured.

The V1.3 wrong answer that produced it (ledger W-6) is a real defect with a named cause, not a statistical accident: a needed quantity discarded through the dismissal escape hatch with a circular reason.

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3. The headline: real language, measured

Sealed 300-item sample drawn from GSM8K and ASDiv, disjoint from the traffic pool. Unresolved items are scored as **abstentions**, never dropped.

arm	certified	correct	wrong	corroboration abstained	coverage	wrong-fire
sonnet	209	206	**3**	12	69.7%	1.44%
haiku	38	37	**1**	142	12.7%	**2.63%** ← above gate
clean combined	**247**	**243**	**4**	154	—	**1.62%**
opus *(excluded, B9)*	199	198	1	1	66.3%	0.50%

Against V1.2's **0.2% coverage at zero wrong**, this is 69.7% coverage at 1.44% wrong on the strongest clean arm. The parse/verify split is what changed, and it is the only thing that changed.

The honest qualification, stated up front

The pre-registered gate is *wrong-fire ≤ 2%*. Sonnet clears it on the point estimate; haiku does not. **No arithmetic arm clears it on the interval:**

track	wrong-fire	95% CI (Wilson)
sonnet arm	3/209 = 1.44%	**[0.49%, 4.13%]**
haiku arm	1/38 = 2.63%	**[0.47%, 13.50%]**
clean arms combined	4/247 = 1.62%	**[0.63%, 4.09%]**
W6 retrieval, both corpora	0/289 = 0.00%	**[0.00%, 1.31%]**

Only the retrieval track clears 2% with confidence. Every arithmetic interval reaches past 4%, and the haiku interval spans an order of magnitude. A version reporting "under the 2% gate" without these columns would be overstating measurements of 209 and 38 samples. Closing the intervals needs more sealed items, not better prose — and that is the single most valuable cheap experiment available to V1.4.

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4. Hypotheses

	claim	predicted	result
L1	traffic recurs in structure	uncertain	**CONFIRMED** — 83.3% at medium granularity vs a 35% gate
L2	frames improve precision, give a no-frame signal	pass	**PARTIAL** — 95.2% recall, 95.6% prompt cut, but 100% no-frame on real language → Amendment A

	claim	predicted	result
L3	parse/verify lifts coverage, wrong-fire ≤ 2%	pass	**SPLIT** — coverage lift emphatic (0.2% → 69.7%); the gate holds on sonnet (1.44%) and **fails on haiku (2.63%)**, which sets the outcome
L4	checks catch misreadings without rejecting good parses	pass	**PARTIAL** — they catch; they also reject good parses (B10)
L5	plan reflexes compound	uncertain	**FALSIFIED** — 7 served, 7 wrong (§6)
L6	a cheap parser suffices	uncertain	**PARTIAL, well short of the pre-registered comparison** — llama3.1:8b, n=22: 7 certified, 7 correct, **0 wrong**, 31.8% coverage vs sonnet's 69.7%. A ~38pp gap on 22 items is not the ±10pp verdict the spec asked for, and no claim is made from it
L7	R3 beats R1; R2 forfeits certification	pass	**PASS, and stronger than predicted** (§5)
L8	provenance ≤ 2% wrong-fire, staleness + conflict caught	pass	**PASS** — 0/289, both hazard classes caught
L9	PARITY-PLUS, Tier A intact	pass	**PASS** — battery, 65-seed sweep, vocab audit, invariants
L10	parse agreement is a useful confidence signal	pass	**PASS, emphatically** — the single largest discriminator (D7)
L11	domain familiarity is measurable	uncertain	**CONFIRMED, in an unexpected direction** (§5)

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5. Not just arithmetic: retrieval certified on provenance

Two hand-built corpora, 49 documents and 300 ground-truthed queries each, identical query design so the only variable is domain. Half the queries are FACTUAL — retrieval only, no computation. Three hazards are planted identically in both: **conflict**, **stale**, **superseded**.

A retrieved fact has no derivation to check, so the guarantee moves onto provenance: *identity → validity window → supersession → source agreement*, and **two live sources that disagree abstain** rather than vote.

corpus	served	correct	wrong	retrieval recall
generic (equipment, tariffs, invoices)	149/150	149	**0**	0.873
user (plastics quoting)	140/150	140	**0**	0.977

These are the first certified non-arithmetic answers in the arc.

289 served, zero wrong, on questions that involve no computation whatsoever.

R1 vs R2 vs R3 — and the middle option that fails

Same queries, same retrieval, three ways to answer (n=120 per corpus, haiku):

arm	what it is	served	wrong	wrong rate
R1	RAG alone: model retrieves and computes	114	42	36.8%
R2	**certified values injected into context**, model still answers	82	47	**57.3%**
R3	engine's value *is* the answer	117	**0**	**0.0%**

§9 predicted R2 would improve accuracy but forfeit certification. It forfeits certification and **does not improve accuracy** — on both corpora it is *worse than plain RAG*. Handing a model verified numbers and asking it to write the answer does not transfer the guarantee; it adds context the model then mis-uses. One servicing agent showed the mechanism directly, reporting that it had *"used supplementary record duty_h value (6.48 hours) for Chiller CU-103 when equipment spec showed 4 hours"* — it picked the planted conflicting source. That is the exact item R3 abstains on.

The guarantee is a property of who computes the answer, not of what is in the context window.

Cross-corpus (L11)

The user's own domain retrieves **better** (0.977 vs 0.873) but serves **less** (140 vs 149). Both follow from one property: plastics records carry distinctive identifiers (`Part PN-3010`, `ABS GP-500`) that hashed n-grams separate cleanly, while the generic corpus is full of colliding names (`Tariff Northland S0`). Better identity means retrieval finds the right record — and also means the engine more often notices it has two genuine sources and abstains. Domain familiarity did not raise accuracy (already perfect on what was served); it moved the loss from *wrong-record* to *honest abstention*. That is the right direction, and it was not predicted.

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6. Self-generating: plan reflexes, falsified

203 verified plans were preloaded into the cache and replayed against traffic they had never seen.

mode	n	served	correct	wrong	hit rate
repeat (with replacement)	3000	6	0	**6**	0.20%
unique (without)	1180	1	0	**1**	0.08%

Seven served, seven wrong. Not a low yield with a small error rate — a 100% wrong-fire rate on everything the cache dared answer.

```
stored    "Mrs. Hilt picked up a book that has 17 pages.
           She read 11 of them..."          sub(count, count)
traffic   "Janet bought 2 chocolate, 13 vanilla and 2 strawberry
           cupcakes. How many cupcakes?"
similarity 0.642 → bound positionally → served 11    truth 17
```

Two short questions about small counts, lexically similar enough to cross threshold, semantically opposite: one subtracts, the other adds.

Every check passed. The values really are in the text, really are all consumed, the plan really does converge on one terminal, `count − count` really is dimensionally legal, the result really is not degenerate.

> ***The checks verify that a plan is internally consistent with the text. Not one of them verifies that the plan answers the question.***

That gap was invisible for eleven versions because a parser always produced the plan from the question in front of it — the plan was about the right question *by construction*. A cache breaks that, and nothing downstream notices. This is V0.1's founding failure, reached again through modern machinery.

L1 still stands: the structures **do** recur, at 83.3%. What does not follow is that a recurring structure can be *bound* to a new question by text similarity. **Structural recurrence is not interchangeability.** The 83.3% is real and unusable by this mechanism. Plan reflexes are not served in V1.3's reported configuration; the measurement is the result.

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7. What the wrong answers were

Five events from the model-parsed path on sealed real language, in three classes. Full detail in `wrong_answer_ledger_v1_3.md`.

Two are non-integer answers under an integral target (both sonnet):

```
"9 boys ... 29 trees ... each watered the equal amount"
    served 3.222   gold 3      (needs FLOOR)
"Jane arranges 16 vases a day. 248 vases. How many days?"
    served 15.5    gold 16     (needs CEILING)
```

Both declare `target.kind = count`. Both serve a non-integer as a count. They need **opposite** roundings, which is why the remedy is abstention, not rounding.

Two are the same English idiom, replicated across two models:

```
"After scoring 14 points, Erin now has three times more points
 than Sara, who scored 8. How many points did Erin have before?"
    opus    served 10.0   corroborator 10.0
    sonnet  served 10.0   corroborator 10.0     gold 18.0
```

Four independent samples across two model families, all four reading *"three times more"* as 3×; the dataset reads it as 4×.

One is a vacuous dismissal — the answer that set the outcome:

```
"Kaleb won eight tickets ... and seven tickets ... candy cost
 five tickets a piece, how many could he buy?"
    plan       add(8, 7) = 15               gold 3
    dismissed  {"value": 5, "reason": "not used in primary
                calculation"}
```

The plan computes the ticket total and never divides by the price. The number it needed sits in `dismissed`, discarded with a reason that is circular — it says the five is unused because it is unused. The escape hatch that lets a parser honestly set aside a distractor was used to set aside the operative quantity, and consumption passed *legitimately*, because the number was accounted for exactly as the schema permits. Nothing in the engine scores the QUALITY of a dismissal, and nothing checks that the declared `target.scope` ("total amount") matches what the question asks.

The finding that matters most

All five share a signature: every deterministic check passes correctly, **and the corroborator agrees with the plan to the digit.** D6 concluded that what closed the wrong-fire gap was a second *opinion*, and that the corroborator "does not need to be right, only independent". This is the boundary of that claim. The corroborator was independent in the only sense the design gave it — different sample, different seed, different prompt shape — and not independent in the sense that mattered. **Two samples of one model share an

interpretation prior.** Resampling varies the plan; it does not vary the reading of the sentence.

The arc now has three statements, each narrowing the last:

```
V1.1   dimensional gates certify physics
V1.2   ...not story semantics
V1.3   ...and resampling does not recover semantics either,
       because the second sample inherits the first one's reading
```

Corroboration removed the *random* errors — 149 abstentions across the clean arms — and left the *systematic* ones untouched.

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8. Safety: two tiers, and a certificate that refuses

`certify.py` now emits both tiers and **fails the whole certificate if Tier A regresses**.

TIER A — self-parsed input. Re-verified live this run: full sealed battery, 10,160 trap items across 65 independent seeds, vocabulary audit, invariants J1–J8. All pass. **Unchanged, unqualified, and regenerable from frozen artifacts.**

TIER B — model-parsed input. Measured, expiring, and refused when no current measurement exists. During the run the certificate printed:

> *"This certificate REFUSES to state a Tier B guarantee because no current measurement exists"* — followed by the excluded evidence, itemised.*

That refusal is the mechanism working, not a failure to produce output. A deployment admitting free-form language runs Tier B and inherits Tier B's rate — not Tier A's guarantee.

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9. Bugs found (15) — the ones that change how to read the numbers

Full detail in `bug_audit_v1_3.md`. Five materially affect interpretation:

- **B7 — bridge token counts are fabricated.** One servicing agent reverse-engineered an *earlier agent's invented formula* and reproduced it faithfully, so the fiction propagated across models and grew more consistent as it spread. **No V1.3 energy or cost figure comes from bridge arms.** This is V1.1's energy contamination recurring through a new channel, caught before it reached a headline rather than after.
- **B9 — a servicing agent read the verifier.** It ran every plan

through `planir.py` before answering. That is a search over the check suite, not a model reading. **The opus arm is excluded from all cross-model claims, and V1.1's "opus mints most" finding is not replicated here.** The accidental upside: even with full sight of the checker it could not make three plans pass.

- **B8 — provenance filters are not commutative.** Supersession ran before conflict detection and resolved planted contradictions by *document order*, serving `22.92` cavities with a full certificate. Ordering is now identity → validity → supersession → conflict.
- **B11 — stubs wearing a refusal.** Agents that ran out of budget emitted `{"refuse": "cannot parse"}`, which is valid JSON and passed ingest. 96 of them inflated haiku's refusal rate from ~0.3% to 11%. Purged and re-run; four further stub shapes (`""`, `{}`, `UNKNOWN`, non-numeric answers to numeric prompts) were closed as they appeared.
- **B12 / B16 — a three-link chain where each fix caused the next.** B5 added word-numbers so "ninety-one hotdogs" could parse at all; that fed phantoms ("the third team" → 0.333) to `check_consumption`, so B12 restricted consumption to digits; which made consumption see *nothing* on a problem written entirely in words, so a plan could declare a quantity, never use it, never dismiss it, and nothing would object. B16 closes it plan-internally — a DECLARED quantity must be used or dismissed, independent of how the text is written. Replaying all three arms from the frozen cache after B16 produced **byte-identical results**, because the wrong answer that exposed the chain went through a different hole (§7). Pre-fix arms preserved in `results/v13/w2/superseded_pre_B16/`.

B5 deserves its own line: spelled-out numbers were unsolvable *by construction*. The checks saw only digits, so a model that correctly read "ninety-one" as 91 failed round-trip, and one that copied the words failed validation. Not a difficulty — a self-inflicted impossibility, invisible as a puzzling rejection rate.

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10. The coverage ceiling is the IR, not the parser

Once B11's stubs are removed, the surviving refusals classify tightly. Roughly one problem in nine is declined, always for one of six structural reasons:

```
GCD / LCM                    not expressible in add/sub/mul/div
remainder / modulo / floor   no such operator
missing world constant       7 days, 4 legs, 100 for a percentage
multi-output answers         "find the length AND the width"
conditional logic            "an extra point if over 20"
clock / calendar arithmetic  wraparound, AM/PM, days in a month
```

None is a reading failure. The model understood every one; the **intermediate representation cannot hold the answer**. V1.3 was built on the premise that the parser was the bottleneck. That premise is now half right: the parser was the bottleneck for problems the IR can express, and for ~11% of real word problems the IR is the wall.

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11. What V1.3 does not claim

- **No energy or cost figure.** The models ran on a subscription through the bridge, and bridge token counts are fabricated (B7). The local arm is the only source of real token counts — it reports 40,527 genuine tokens over 22 items — but no joule meter was run against it, so V1.3 states token counts and **no energy figure at all**. L6, the commercial crux, is answered only directionally: the cheap local parser certified 7 of 22 at zero wrong, roughly 38 points of coverage behind sonnet. That is a signal, not the ±10pp verdict the spec asked for.
- **No cross-model ranking.** One of three arms is disqualified (B9); two clean arms are not a ranking.
- **No claim that plan reflexes work.** They do not (§6).
- **No claim that the 2% gate is cleared with confidence** on the arithmetic track (§3).
- **26 items across the arms remain unresolved** and are counted as abstentions — the conservative direction, which can understate coverage but never overstate it.

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12. V1.4 worklist, ordered by evidence

1. **Check 8, TARGET INTEGRALITY.** If the target kind is integral, the value must be an integer, else abstain — never round. Would have caught **2 of the 3** clean-arm wrong answers, taking sonnet from 1.44% to 0.58%. Deterministic, model-free, no extra call. Its coverage cost is unmeasured and must be measured before adoption. *Deliberately not implemented in V1.3: adding it after

seeing the sealed results would be tuning on the evaluation.*

2. **Extend the plan IR** with `mod`, `gcd`, `lcm`, `floordiv` and a provenance-carrying constant table. Five of the six refusal classes are new operators, each dimensionally checkable the way the arc's existing gates expect. Addresses ~11% of real problems.
3. **Cross-model corroboration — re-scoped downward.** D11 proposed it as the headline fix; the W-1 replication across two model families weakens that. A different model helps only where models differ, and on common English idioms they do not. Worth testing; no longer first.
4. **Re-found or drop plan reflexes.** As specified the design is circular: a safe match needs the question's semantics, which is the parse the cache exists to avoid. Either drop it, or re-found it on a cheap pre-parse signal (the target-kind and operator signature the frame could predict) with **wrong-fire measured before hit rate**.
5. **A real energy measurement**, on hardware that will not time out.
6. **Ambiguous-quantifier detection** ("times more", "as much again") forcing abstention — a candidate mitigation for W-1 with a measurable coverage cost.
7. **Score the dismissal, and the scope.** A dismissal must cite a span and a role rather than assert non-use, and dismissing a quantity introduced by the question's final clause should be refused outright. This is the W-6 class — the one that set the version's outcome — and it is deterministic and cheap.
8. **More sealed items.** Every arithmetic interval in §3 reaches past 4% because n is 209 and 38. No engine change improves that; only more sealed evaluation does, and it is the cheapest thing on this list.

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13. Artifacts

- `V1_3_REPORT.pdf` — this report, 12 pages
- `V1_3_SPEC.md` — the frozen spec, with Amendments A and B
- `bug_audit_v1_3.md` — 16 bugs, each with what it cost
- `invention_log_v1_3.md` — discoveries D1–D12
- `wrong_answer_ledger_v1_3.md` — every wrong answer, nothing excused
- Web summary: https://claude.ai/code/artifact/6c7ab461-58c4-48eb-b9ce-633bdff76540
- Tag `v1.3-final`; snapshot `reflex-engine-v1.3-frozen` (9,723 files,

`.env` excluded)

13b. Reproduction

```
python run_v1_3_experiment.py
```

Reproduces exactly, offline and with no model: W0 recurrence, W1 frame, both corpora, W6 R3 on both, W5, Tier A certification, and the aggregate table. The W2 parse arms replay **byte-for-byte from the frozen response cache** in `results/bridge_v13/` — reproduction of the measurement, not of the model, and the distinction is stated rather than blurred.

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14. The one-sentence version

The engine now reads ordinary English and answers 69.7% of it with three wrong answers in 209, and separately certifies 289 retrieved facts with none wrong — but it missed its own safety gate on the weakest arm it measured, which is why this is an outcome D and not an outcome A; the two mechanisms meant to make it cheap and self-improving turned out to be unmeasured and false respectively; and the most valuable thing learned is that a model's second opinion about a sentence is not independent of its first.